Ths Pg Hs N Vwls

i ae a o oe

Three Quantum Computing Tapas

Tonight's Menu

adiabatic quantum computer..... $√money in your wallet
or $∞?

topological quantum computation..... ~$109

quantum computer with only measurements..... $0.37

Chez Quantum

Dave Bacon

"After Bugs Bunny"

Institute for Quantum Information

California Institute of Technology

Novel Models for Quantum Computation

or "Why is Dave I a physics bigot?"

What it *means* to compute is dictated by physics.

---

1. Physicists have *tools* for studying quantum systems.

   Adiabatic theorem of QM $\Leftrightarrow$ Adiabatic quantum computation.

Theoretical condensed matter $\Leftrightarrow$ Topological quantum computation

---

2. Physics dictates how we count resources.

   What is the power of an adiabatic quantum computer?

   What resources are needed for universal quantum computation?

---

3. What is the *physics* of robust computation?

   Naturally fault-tolerant quantum computation ■ Topological quantum computation

---

Inspirations for new directions in quantum computation.

Adiabatic Quantum Computation

Hamiltonians

The mystery of where "gates" come from:

$$i\hbar\frac{d|\psi(t)\rangle}{dt} = \mathbf{H}|\psi(t)\rangle$$

Hamiltonian

Operator on $\mathcal{H}$, $|\psi\rangle \in \mathcal{H}$

Schrödinger equation

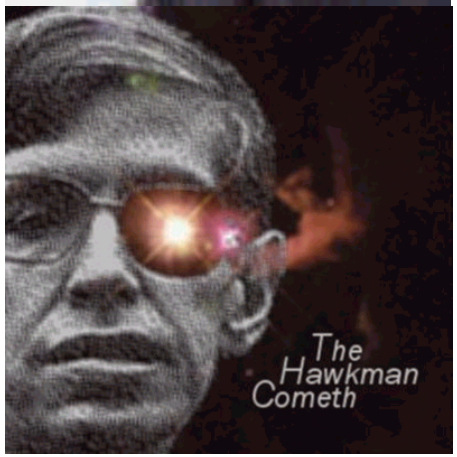
$$|\psi(t)\rangle = \mathbf{U}(t)|\psi(0)\rangle$$

$\mathbf{H}$ independent of t $\mathbf{U}(t) = \exp\left[-\frac{i\mathbf{H}t}{\hbar}\right]$
$\mathbf{H}$ depends on t $i\hbar\frac{d}{dt}\mathbf{U}(t) = \mathbf{H}(t)\mathbf{U}(t)$

$$\mathbf{U}(t) = \mathbf{I} + \frac{1}{i\hbar}\int_0^T \mathbf{H}(t')dt' + \frac{1}{(i\hbar)^2}\int_0^T\int_0^T \mathbf{H}(t'')\mathbf{H}(t')dt'dt'' + \cdots$$

$\hbar = 1$ from now on.





Stephen Hawking

Hamiltonian
=
Ham and melted
cheese sandwich

NP
=
Not Polynomial



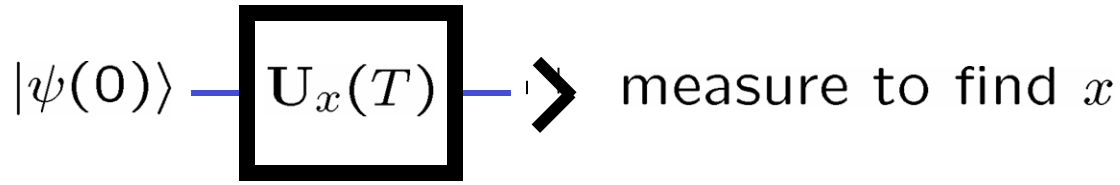
Alan Turing

Precursor

A "Continuous-time" Search Algorithm

Given: ability to perform $\mathbf{H}_x = |x\rangle\langle x|,\ x \in \{0, \ldots, N\}$

Task: find x



$$\mathbf{U}_x(t) = \exp\left[-i|x\rangle\langle x|t\right] = e^{-it}|x\rangle\langle x| + \textstyle\sum_{y\neq x} |y\rangle\langle y|$$

Initial state $|\psi(0)\rangle = \sum_y \alpha_y |y\rangle$

$$|\psi_x(T)\rangle = \alpha_x e^{-it}|x\rangle + \textstyle\sum_{y\neq x} \alpha_y |y\rangle$$

$$\min|\langle\psi_x(T)|\psi'_x(T)\rangle| = \min\left(|\alpha_x|^2 e^{iT} + |\alpha_{x'}|^2 e^{-iT} + \textstyle\sum_{y\neq x,x'} |\alpha_y|^2\right)$$

One T for all $\mathbf{H}_x \rightarrow |\alpha_x| = |\alpha| = \frac{1}{\sqrt{N}}$. min at $T = \pi$

$$\min|\langle\psi_x(T)|\psi'_x(T)\rangle|^2 = \left(1 - \tfrac{4}{N}\right)^2$$

Must make $O(N)$ measurements to distinguish different x's.

Total time $= O(N)$

Drivers License

Given: ability to perform $\mathbf{H}_x = |x\rangle\langle x|,\ x \in \{0, \ldots, N\}$
as well as a driving Hamiltonian $\mathbf{H}_D$

$$\mathbf{H}'_x = \mathbf{H}_x + \mathbf{H}_D$$

Task: find x

$$\mathbf{H}_D = |\phi\rangle\langle\phi|,\ |\phi\rangle = \tfrac{1}{\sqrt{N}} \textstyle\sum_y |y\rangle$$

$\mathbf{H}'_x$ acts on subspace spanned by $|x\rangle$ and $|x^\perp\rangle = \frac{1}{\sqrt{N-1}} \sum_{y \neq x} |y\rangle$

$$\mathbf{H}_x = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \mathbf{H}_D = \begin{pmatrix} \frac{1}{N} & \frac{\sqrt{N-1}}{N} \\ \frac{\sqrt{N-1}}{N} & \frac{N-1}{N} \end{pmatrix}$$

$$\mathbf{H}'_x = \mathbf{I} + \tfrac{1}{N}\mathbf{Z} + \tfrac{\sqrt{N-1}}{N}\mathbf{X}$$

$$\mathbf{U}_x(t) = e^{-it}\left(\cos\left(\tfrac{t}{\sqrt{N}}\right)\mathbf{I} - i\sin\left(\tfrac{t}{\sqrt{N}}\right)\left(\tfrac{1}{\sqrt{N}}\mathbf{Z} + \sqrt{1-\tfrac{1}{N}}\mathbf{X}\right)\right)$$

$$\mathbf{U}\left(\tfrac{\pi}{2}\sqrt{N}\right)|\phi\rangle = |x\rangle \qquad |\phi\rangle = \tfrac{1}{\sqrt{N}}|x\rangle + \sqrt{1-\tfrac{1}{N}}|x^\perp\rangle$$

Total time $= O(\sqrt{N})$

Grover speedup!

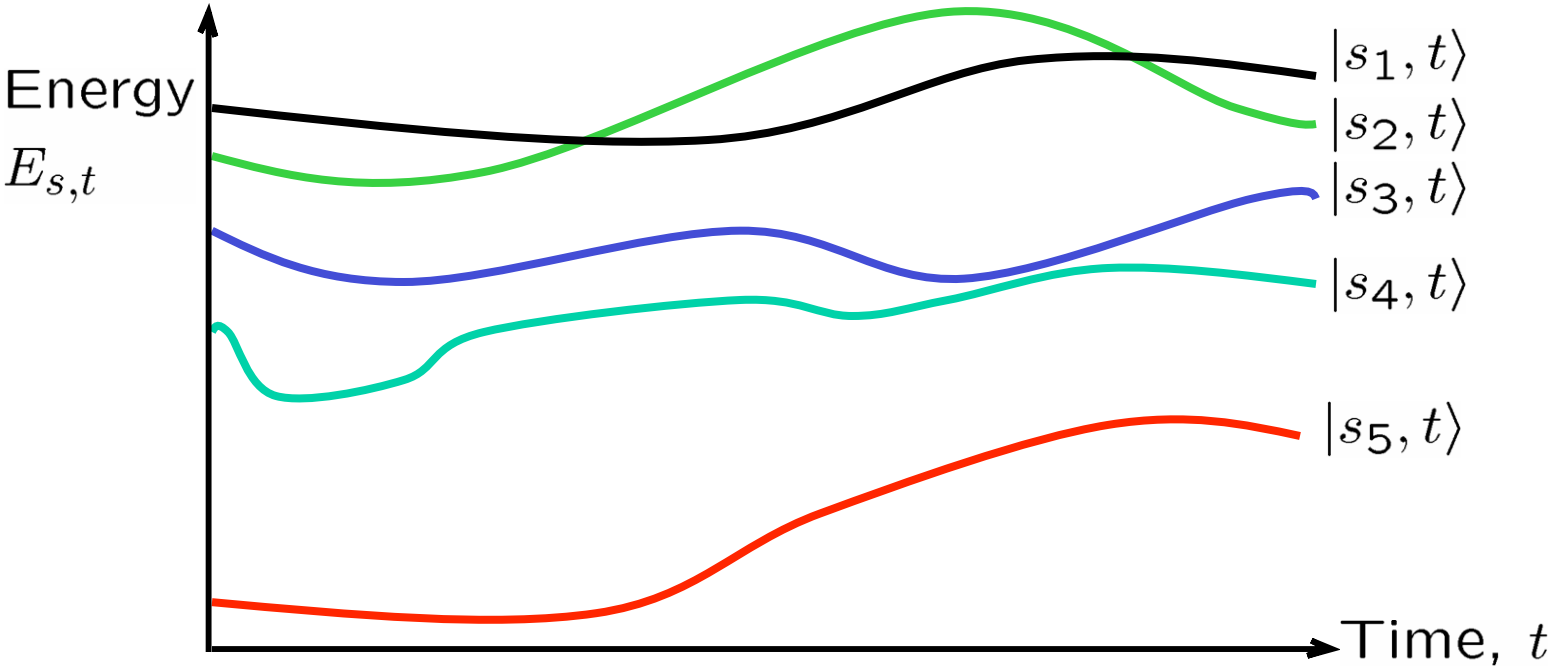
The Adiabatic Theorem

Time dependent Hamiltonina $\mathbf{H}(t)$

Instanteneous eigenstates $\mathbf{H}(t)|s,t\rangle = E_{s,t}|s,t\rangle$

At $t = 0$ prepare system into eigenstate of $\mathbf{H}(t = 0)$, $|s_0, 0\rangle$

If $\mathbf{H}(t)$ varies "slow enough" and the eigenstate does not cross other eigenstates then evolution nearly perfectly follows the eigenstate.



What Does Slow Mean?

Adiabatic condition

If $\frac{d_{max}}{g_{min}^2} \leq \epsilon$, then $|\langle s_0, T|\psi(T)\rangle|^2 \geq 1 - \epsilon^2$

$g_{min} = \min_{s_i \neq s_0, 0 \leq t \leq T} |E_{s_0,t} - E_{s_i,t}|$ = minimum energy gap

$d_{max} = \max_{0 \leq t \leq T} \left|\langle s_0, 0|\frac{d\mathbf{H}(t)}{dt}|s_0, T\rangle\right|$

large gap $\Rightarrow$ large "success" probability

Adiabatic Algorithms

Suppose

Measuring lowest eigenstates of $\mathbf{H}(T)$ produces the solution to an algorithmic problem of size n.

We can efficiently prepare lowest eigenstate of $\mathbf{H}(0)$.

We can efficiently implement evolution due to $\mathbf{H}(t)$

$\frac{d_{max}}{g_{min}^2} \leq \epsilon = \frac{1}{O(n^k)}$

Then

Input lowest eigenstate of $\mathbf{H}(0)$

Evolve system according to $\mathbf{H}(t)$

Measure output state.

$\Rightarrow$ solves algorithmic problem in time T

Farhi, Goldstone, Gutmann, and Sipser (2000)

Example

$\mathbf{H}_f = \sum_x f(x)|x\rangle\langle x|$ $\quad$ $f(x) : \{0,1\}^n \rightarrow \mathsf{R}$

$f(x)$ is bounded by a polynomial in n.

Find unique minimum of f.

$\mathbf{H}_0 = -|\psi\rangle\langle\psi|$ $\quad$ $|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle$

$$\boxed{\mathbf{H}(t) = (1 - \tfrac{t}{T})\mathbf{H}_0 + \tfrac{t}{T}\mathbf{H}_f} \qquad 0 \leq t \leq T$$

$\frac{d\mathbf{H}(s)}{dt} = \frac{\mathbf{H}_f - \mathbf{H}_0}{T}$ $\quad$ $d_{max} = \max_{0 \leq t \leq T} \left| \langle s_0, 0 | \frac{d\mathbf{H}(t)}{dt} | s_0, T \rangle \right| = \frac{O(n^k)}{T}$

To succeed with probability $1 - \epsilon^2$,

$$\epsilon \geq \frac{O(n^k)}{T g_{min}^2} \;\Rightarrow\; T \geq \frac{O(n^k)}{g_{min}^2 \epsilon}$$

If $g_{min} = O(n^{-l})$, then $T = O(n^m)$.

If $g_{min} = O(c^{-n})$, then $T = O(c^n)$.

What is the Power?

Search for x_0:

$$f(x) = -\delta_{x,x_0}$$

$$\mathbf{H}(t) = -\left(1 - \tfrac{t}{T}\right)|\psi\rangle\langle\psi| - \tfrac{t}{T}|x_0\rangle\langle x_0|$$

Use basis $|x_0\rangle$, $|x_0^{\perp}\rangle = \frac{1}{\sqrt{2^n}}\sum_{x\neq x_0}|x\rangle$

$$\mathbf{H}(t) = -\tfrac{t}{T}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \left(1 - \tfrac{t}{T}\right)\begin{pmatrix} \frac{1}{N} & \frac{\sqrt{N-1}}{N} \\ \frac{\sqrt{N-1}}{N} & \frac{N-1}{N} \end{pmatrix} \qquad N = 2^n$$

$$g^2(t) = 1 - 4\tfrac{2^n-1}{2^n}\tfrac{t}{T}\left(1 - \tfrac{t}{T}\right) \Rightarrow g_{min}^2 = \tfrac{1}{2^n} \qquad d_{max} \leq \tfrac{1}{T}$$

To succeed with probability $1 - \epsilon^2$,

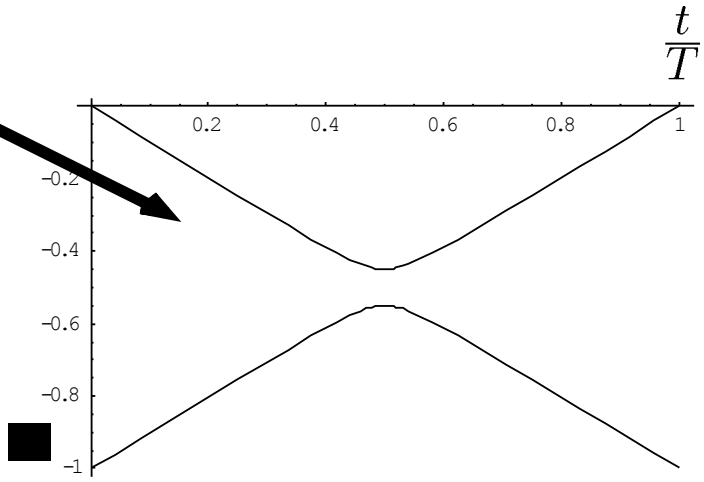
$$\epsilon \geq \frac{2^n}{T} \;\Rightarrow\; T \geq \frac{2^n}{\epsilon}$$

As bad as classical!

The Power of Change

$$\mathbf{H}(t) = -\left(1 - \frac{t}{T}\right)|\psi\rangle\langle\psi| + \frac{t}{T}|x_0\rangle\langle x_0|$$

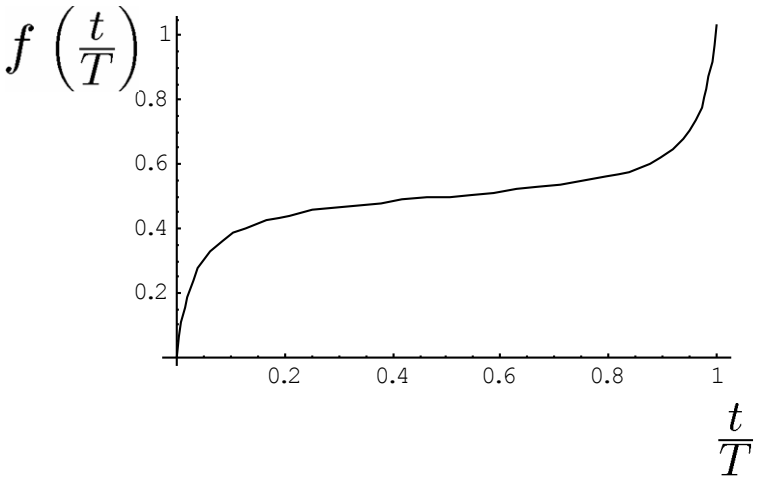
Small region where g_{min} is small.



So spend more time where gap is small?

$$\mathbf{H}(t) = -\left(1 - f\left(\frac{t}{T}\right)\right)|\psi\rangle\langle\psi| - f\left(\frac{t}{T}\right)|x_0\rangle\langle x_0|$$

Takes time $T = O(\sqrt{2^n})$!



We already know this!

Impulsive approximation + continuous grover!

3SAT Adiabatically?

3SAT

Given: a formula $\Phi = C_1 \wedge C_2 \wedge \cdots \wedge C_M$ where C_M are clauses in the variables $x_1, \ldots, x_n$ (i.e.$C_1 = x_1 \vee x_5 \vee \bar{x_8}$)

Problem: Does Φ have a satisfying assignment?

$$h_C(x_{i_C}, x_{j_C}, x_{k_C}) = \begin{cases} 0 \text{ if } x_{i_C}, x_{j_C}, x_{j_C} \text{ satisfies clause } C \\ 1 \text{ otherwise} \end{cases}$$

$$\mathbf{H}_f = \sum_i \mathbf{H}_{C_i} \qquad \mathbf{H}_{C_i}|x_1, \ldots, x_n\rangle = h_C(x_{i_C}, x_{j_C}, x_{k_C})|x_1, \ldots, x_n\rangle$$

Note: $\mathbf{H}_f$ is 3-local.

Lowest eigenvalue of $\mathbf{H}_f$ is zero iff Φ has a satisfying assignment.

$\mathbf{H}_0 = -|\psi\rangle\langle\psi|$
$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle$

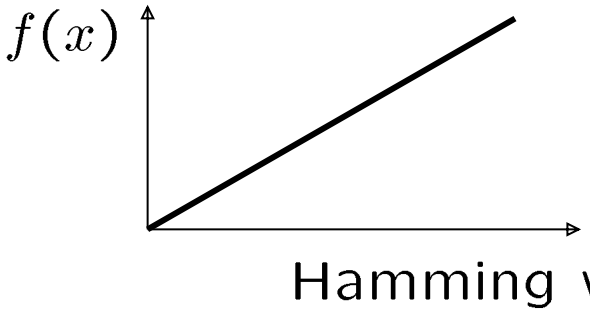
$+ \quad \mathbf{H}_f = \sum_i \mathbf{H}_{C_i}$ — Solution to 3SAT?

Not Yet, Not Ever?

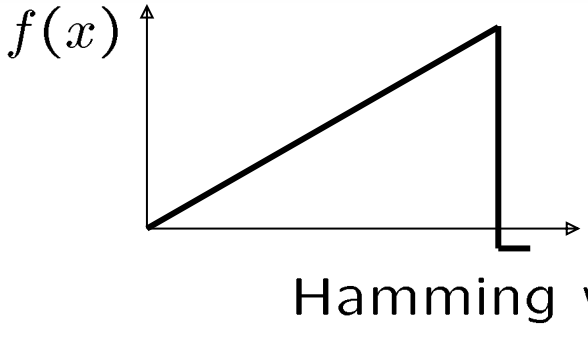
Farhi, Goldstone, Gutmann: "numerical evidence" for polytime.

van Dam, Mosca, Vazirani: black box querry complexity for # clauses oracle does not rule out efficient adiabatic algorithm.

van Dam, Mosca, Vazirani: Perturbed hamming weight problem not solvable efficiently.



Adiabatic Good



Adiabatic Bad

van Dam, Vazirani: bad instance of 3SAT.

Farhi, Goldstone, Gutmann: but different than classical annealing?

Infinite dollar question: exponential separation!

Topological Quantum Computation

Rant Mode On

Damnit!

PARENTAL ADVISORY EXPLICIT LYRICS

Only physicists understand timescales.

EVENTUALLY your hard drive will fail.

EVENTUALLY your PC will produce a hardware error.

Timescale for errors in classical computers are extremely long

Why? Why? Why? Why?

Physics. Physics? Physics. Physics!

Carver Mead Principle of Computation

There are distinct PHYSICAL reasons why classical computers are robust.

Corollary: Not all physical systems are created equal.

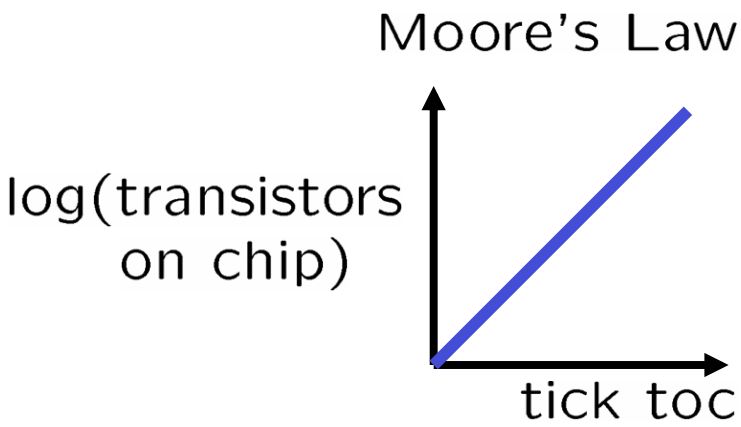
DO NOT FORGET OUR PRESILICON HISTORY

(NOT SO SUCCESSFUL)

Damnit! Part II

Rant Mode On

The computer revolution was (is) about scaling.



Moore's Lucky Guess

Moore's Self-fulfilling Prophecy

Classical computer scalability worked from top towards bottom

One could start from microscopic systems as the basic building blocks, but not technologically easy.

The dogma (meme): "quantum computer", being quantum (duh), must be microscopic.

Reasons to be a heretic: superconductivity, quantum hall effect, Alexi Kitaev's topological quantum codes, (this talk?),...

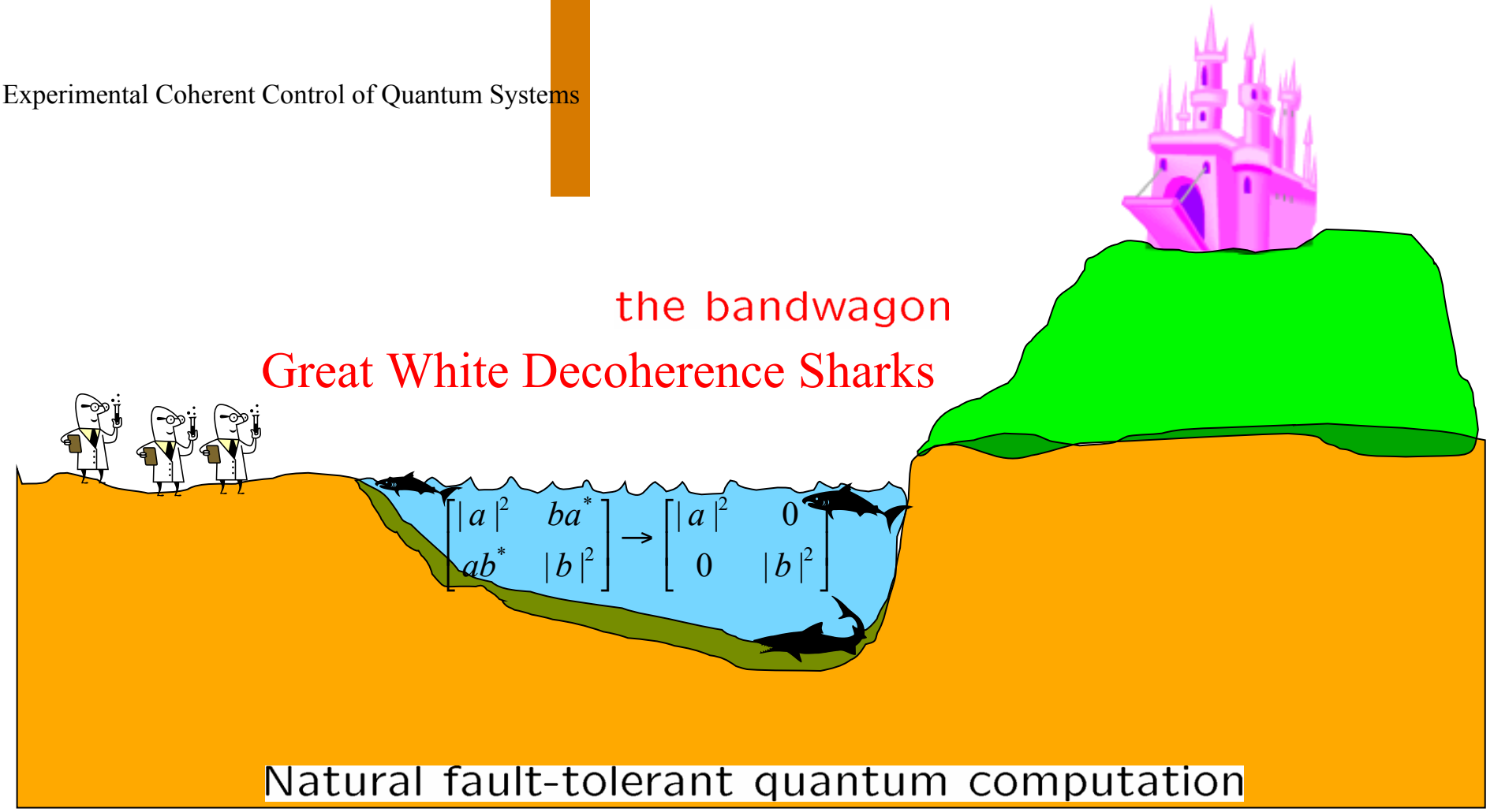
...Lesson of quantum error correction: not too different from classical error correction

Fight the Gap

Rant Mode Shutting Down

Fault-Tolerant Quantum Error Correction

Experimental Coherent Control of Quantum Systems

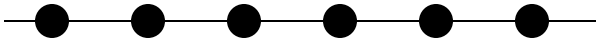


A Lesson

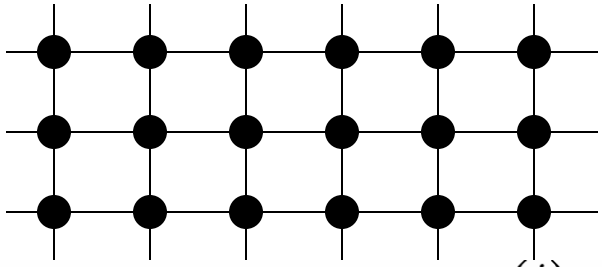
Classical Ising memory

$$\mathbf{H} = -J \sum_{\langle i,j \rangle} \boldsymbol{\sigma}_z^{(i)} \boldsymbol{\sigma}_z^{(j)}$$

One dimensional Ising model



Two dimensional Ising model

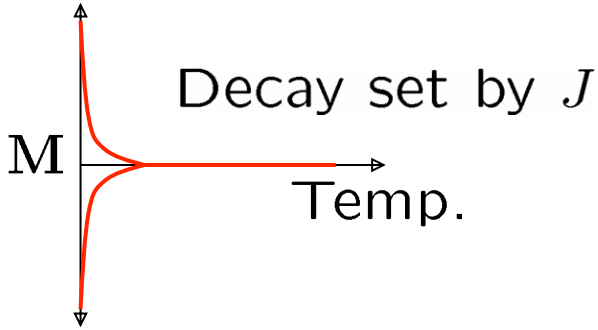


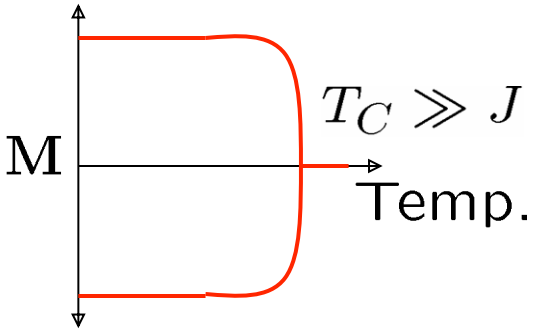
Redundancy code: Stabilizer code with elements $\boldsymbol{\sigma}_z^{(i)} \boldsymbol{\sigma}_z^{(j)}$

Codewords are ground states $|0 \cdots 0\rangle$, $|1 \cdots 1\rangle$

Order parameter: total magnetization, $\mathbf{M} = \sum_i \boldsymbol{\sigma}_z^{(i)}$

indication that the information is "still there"

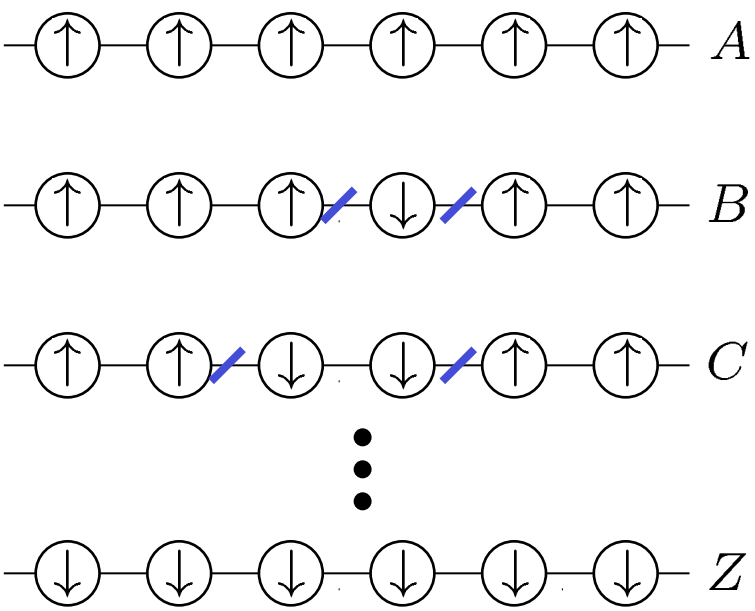


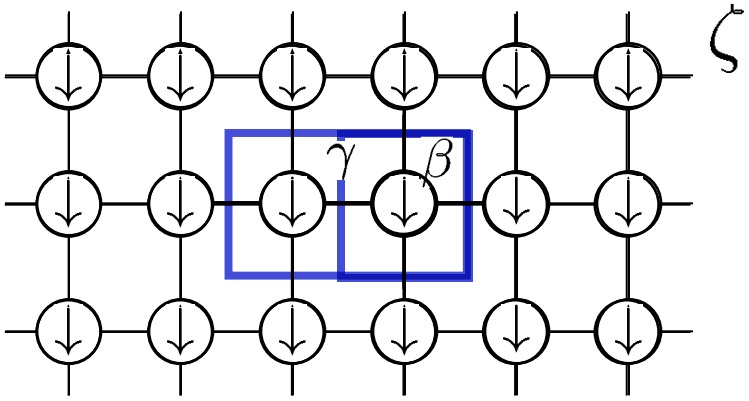


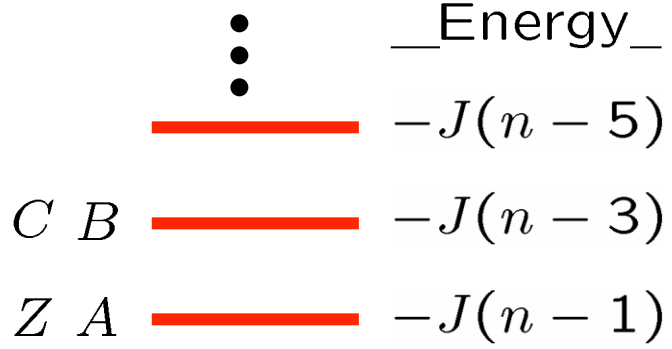
Don't build memory in 1D

2D memories much better

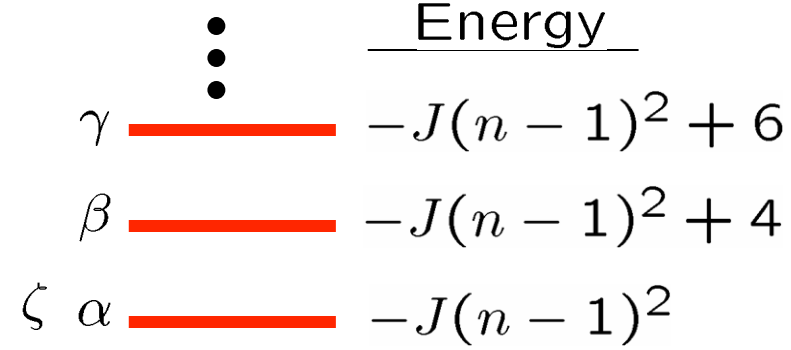
What Have We Learned?





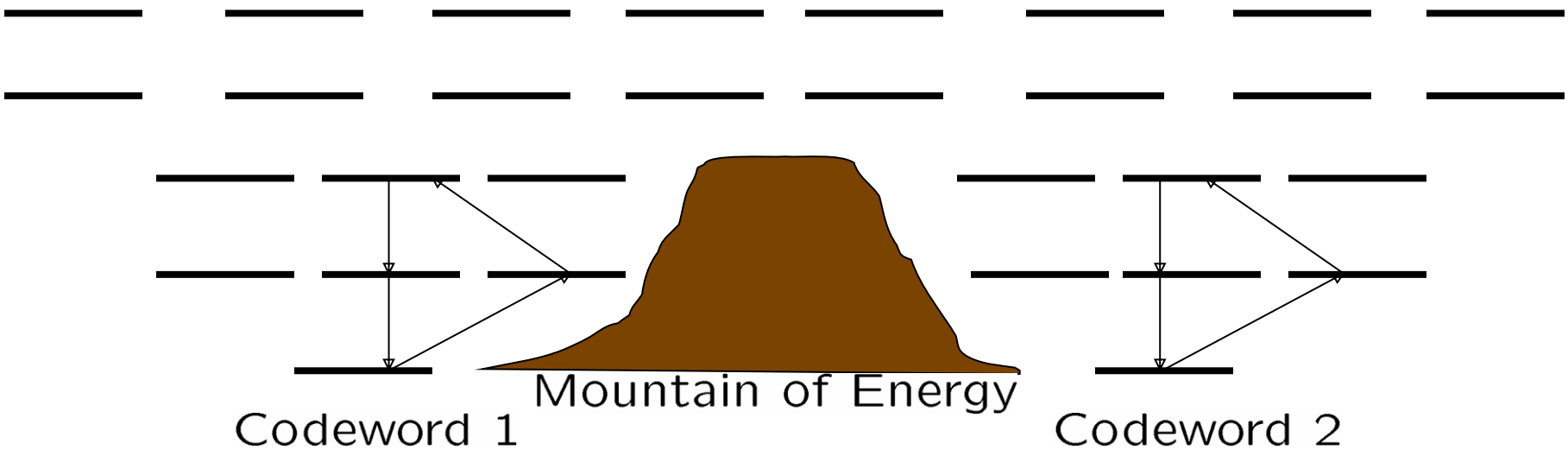


Only $O(J)$ energy used to move between codewords (error)



$O(nJ)$ energy used to move between codewords (error)

Fixing Funnel



Macroscopic expendature of energy to cause errors

Errors are "self-correcting"

Physics does the error correcting

Encoded operation fault-tolerant

Natural Fault Tolerance

Do there exist or can we engineer quantum systems which are naturally fault tolerant?

Become a physicist: think about Hamiltonians $\mathbf{H}$

Three basic requirments

1. Energy levels are error correcting code ~~spaces~~ Subsystems.

   Degeneracy of the energy levels is encoded quinformation

   Symmetry of Hamiltonian

2. Persistence of coherence order parameter at non-zero temperature.

   What is the order parameter for quantum error correction?

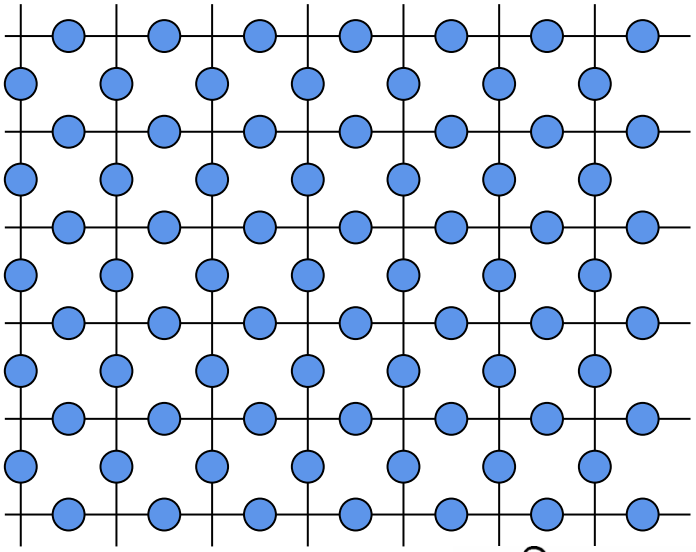
3. Fault-tolerant universal quantum operations

   Macroscopice energy expendature to enact gate

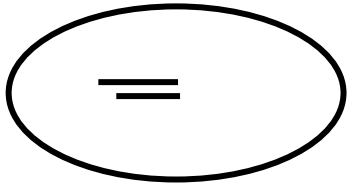
Toric Codes

Kitaev 1997

n by n square lattice on a torus

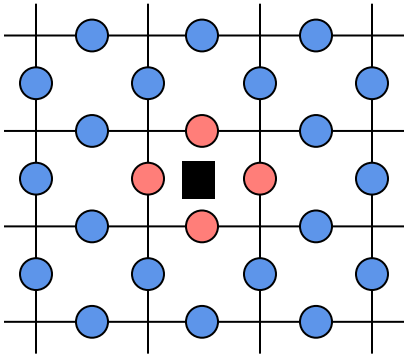


$2n^2$ qubits



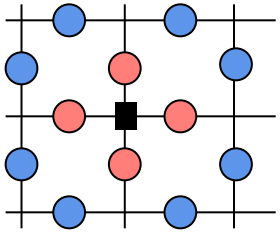
plaque operators

$$\mathbf{B}_p = \mathbf{Z}_1\mathbf{Z}_2\mathbf{Z}_3\mathbf{Z}_4$$



vertex operators

$$\mathbf{A}_s = \mathbf{X}_1\mathbf{X}_2\mathbf{X}_3\mathbf{X}_4$$



$$[\mathbf{X}\otimes\mathbf{X}, \mathbf{Z}\otimes\mathbf{Z}] = 0$$

plaque and vertex operators commute.

plaque and vertex operators have eigenvalues ± 1.

The toric Hamiltonian:

$$\mathbf{H} = -E_0\left[\sum_e \mathbf{A}_e + \sum_p \mathbf{B}_p\right]$$

There's a Code in my Hamiltonian

(…but no Wocket in my Pocket)

$$\mathbf{H} = -E_0 \left[\sum_e \mathbf{A}_e + \sum_p \mathbf{B}_p \right]$$

Since $\mathbf{A}_e$ and $\mathbf{B}_p$ commute we can simultaneously diagonalize these operators.

$$\mathbf{A}_e|\psi\rangle = A_e|\psi\rangle,\ \mathbf{B}_p|\psi\rangle = B_p|\psi\rangle \quad A_e, B_p \in \{+1, -1\}$$

$$|\psi\rangle = |A_1, \ldots, A_n, B_1, \ldots, B_n, \lambda\rangle$$

But $\prod_e \mathbf{A}_e = \mathbf{I}$, so $\prod_e A_e = +1$ $\qquad$ $\prod_p \mathbf{B}_e = \mathbf{I}$, so $\prod_e B_e = +1$

$A_1, \ldots, A_n, B_1, \ldots, B_n$ span Hilbert space of dim $= 2^{2(n-1)}$

$$|\psi\rangle = |A_1, \ldots, A_n, B_1, \ldots, B_n, \lambda\rangle,\ \lambda \in \{00, 01, 10, 11\}$$

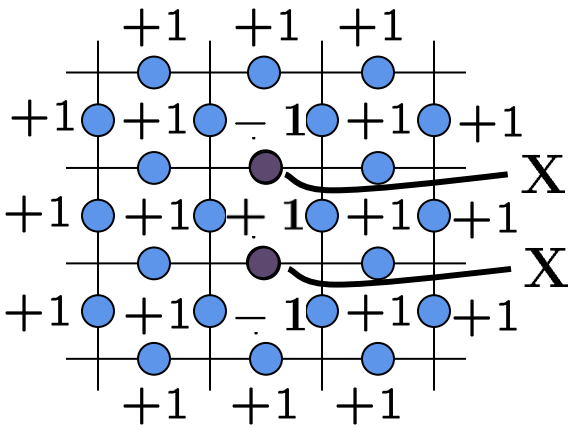
Ground state of $\mathbf{H} = -E_0 \left[\sum_e \mathbf{A}_e + \sum_p \mathbf{B}_p \right]$ has $A_i = B_i = +1$ and is four-fold degenerate (λ).

Anticommuting is Not a Conservative Agenda Item

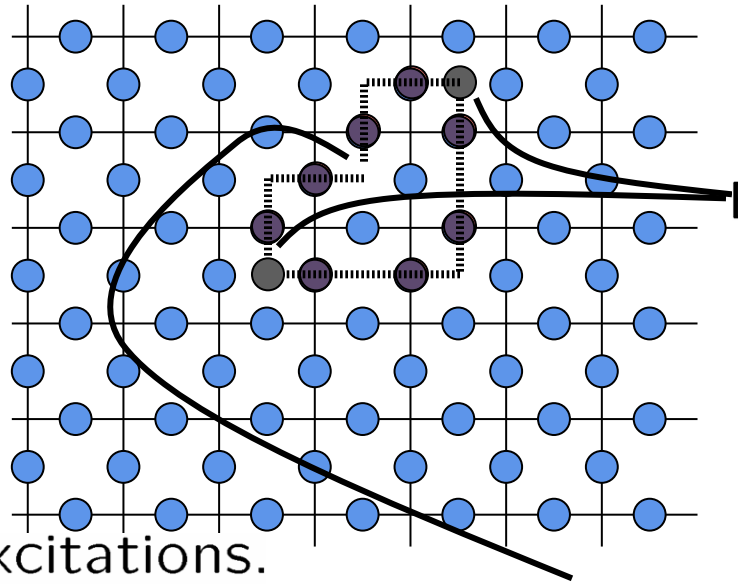
If $\{\mathbf{B}_p, \mathbf{E}\} = \mathbf{B}_p\mathbf{E} + \mathbf{E}\mathbf{B}_p = 0$ and $\mathbf{B}_p|\psi\rangle = B_p|\psi\rangle$, then

$$\mathbf{B}_p(\mathbf{E}|\psi\rangle) = -\mathbf{E}\mathbf{B}_p|\psi\rangle = -B_p(\mathbf{E}|\psi\rangle)$$

flips sign of ± 1 eigenvalue.



$$\mathbf{B}_p = \mathbf{Z}_1\mathbf{Z}_2\mathbf{Z}_3\mathbf{Z}_4$$

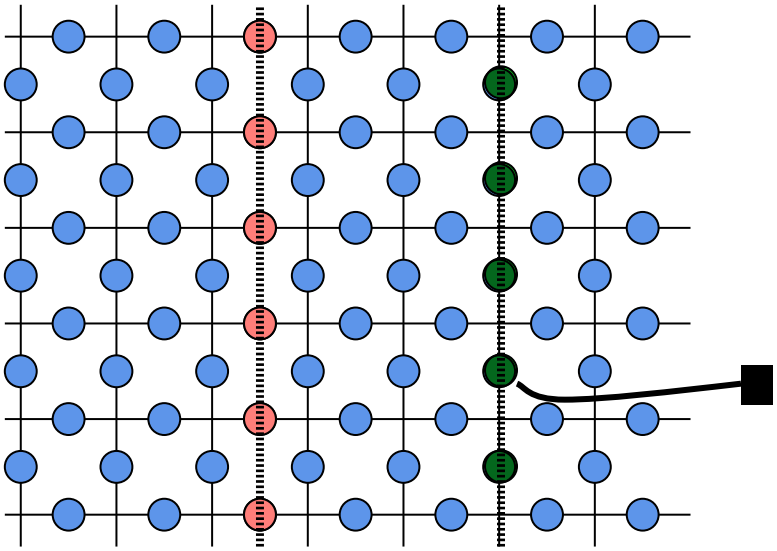


Measure all $\mathbf{B}_p$ to identify excitations.

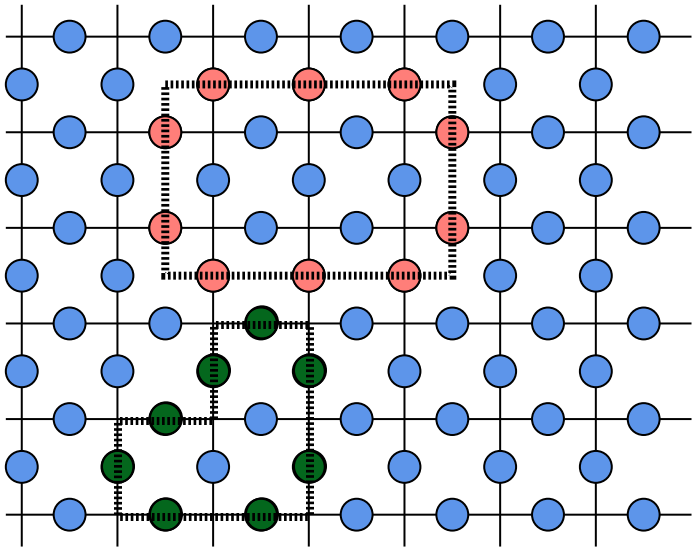
Fix by connecting excitations.

If path is contractable loop then error and fix = product of $\mathbf{A}_e$ s.t. $\prod|\psi\rangle = |\psi\rangle$

Mmmm…Topology



homologically nontrivial



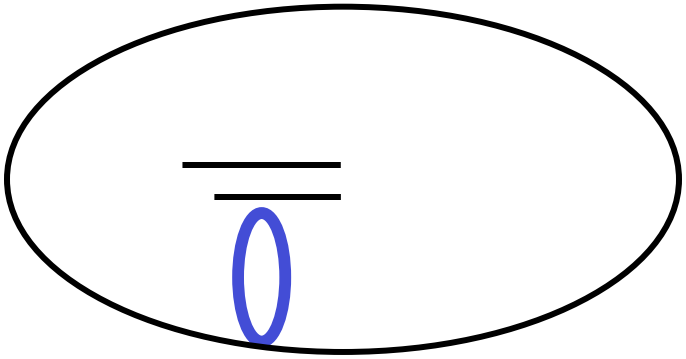
homologically trivial

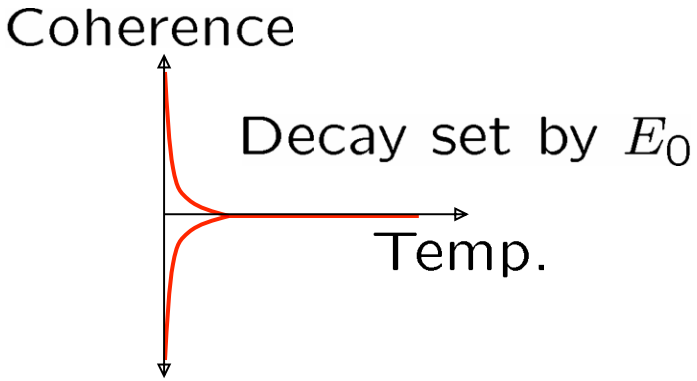
If path is non contractable loop then error and fix $\neq$ product of $\mathbf{A}_e$. Error!

Topological protection of the quantum information.

Where Are We?

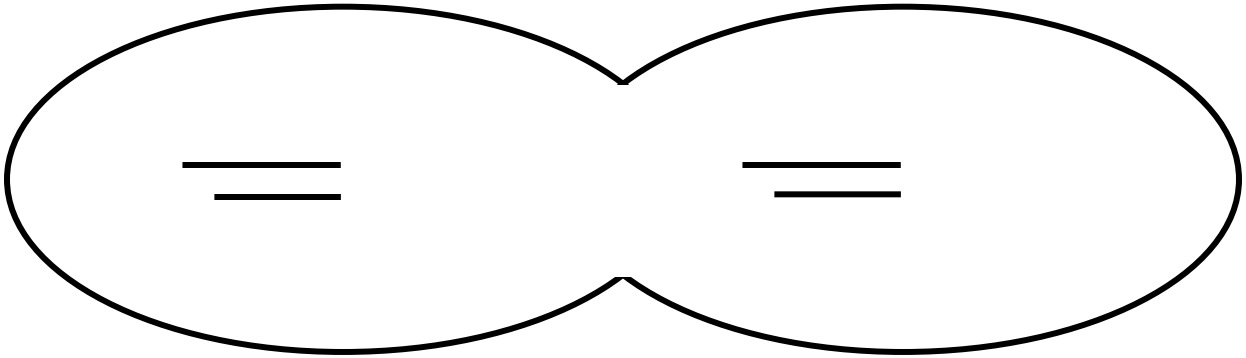
Topological protection of the quantum information.





Virtual tuneling of anyon's creates energy splitting $\propto e^{-cn}$.

Genus g surface has degeneracy 4^g



Room Temperature Quantum Computer?

Problems with toric code:

4 qubit interactions.

4 qubit interactions usually effective interactions and weak.

not like 2D Ising model: excitations are particlelike.

(In 4D toric code, excitations are like 2D Ising.)

Hard to get E_0 large ($>$ few K), even for 2 qubit interactions

Is there a room temperature quantum memory?

Measurement Based Quantum Computation

Two qubit projective measurements via teleportation

Nielsen and Leung

Linear optics plus photon preparation and counting

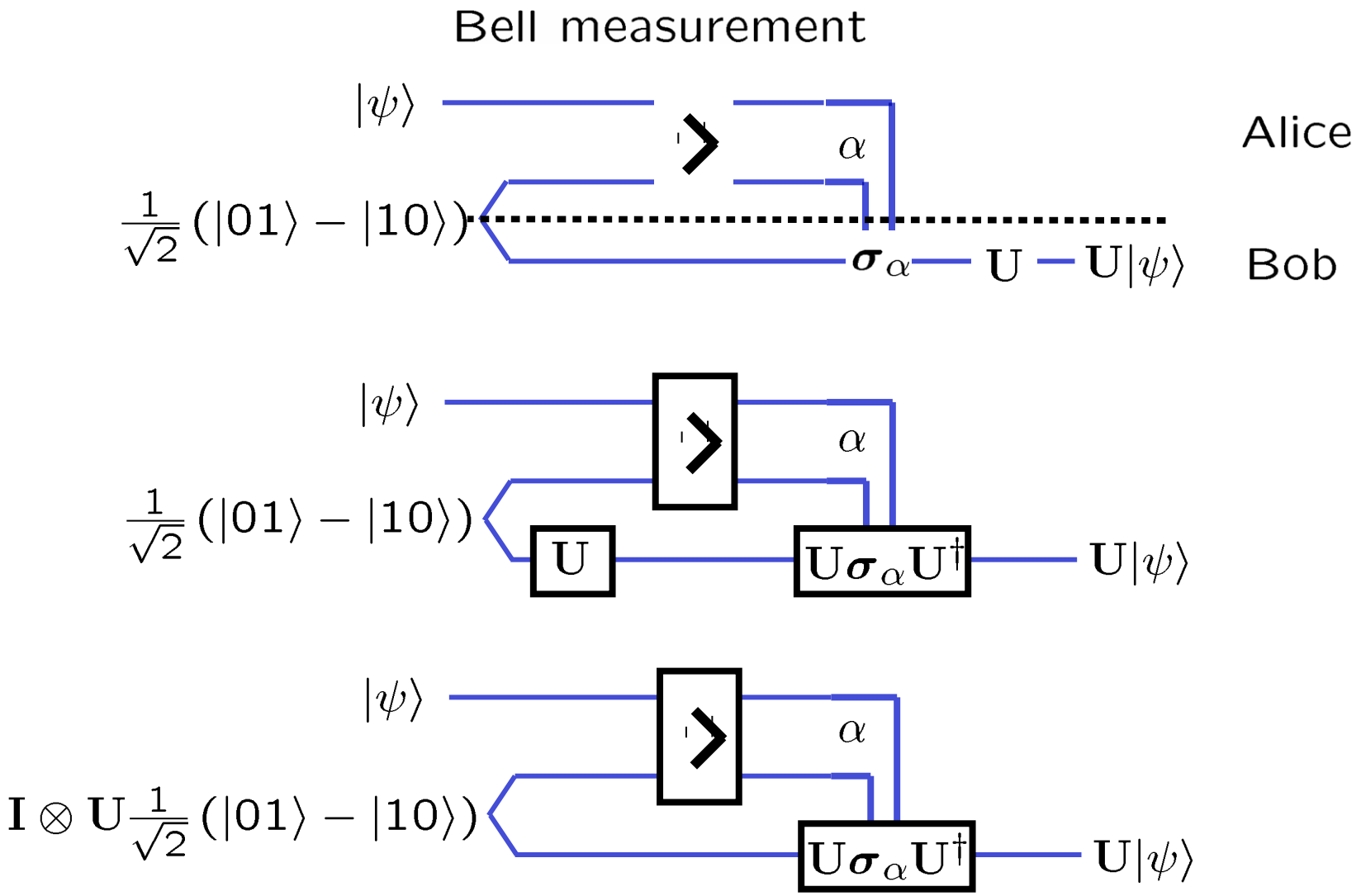
Knill, Laflamme, and Milburn

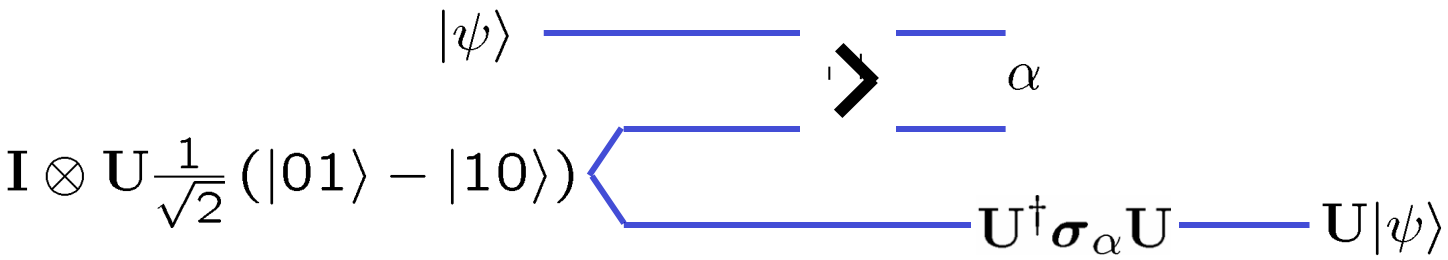
Quantum computing with prior entanglement and measurements

Raussendorf and Briegel

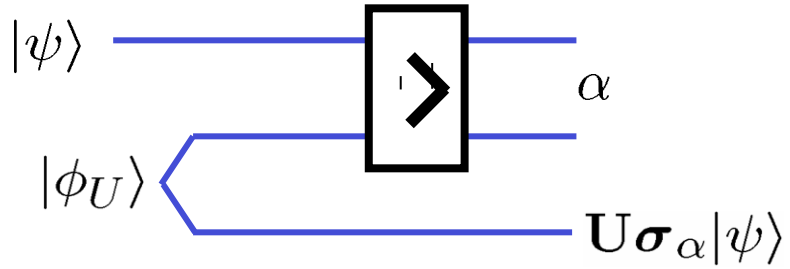
Random Gate

Teleport and apply gate:

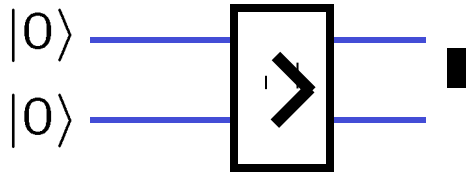




α	Probability	Gate
■	$\frac{1}{4}$	$\mathbf{U}$
■	$\frac{1}{4}$	$\mathbf{U}\sigma_x$
■	$\frac{1}{4}$	$\mathbf{U}\sigma_y$
■	$\frac{1}{4}$	$\mathbf{U}\sigma_z$



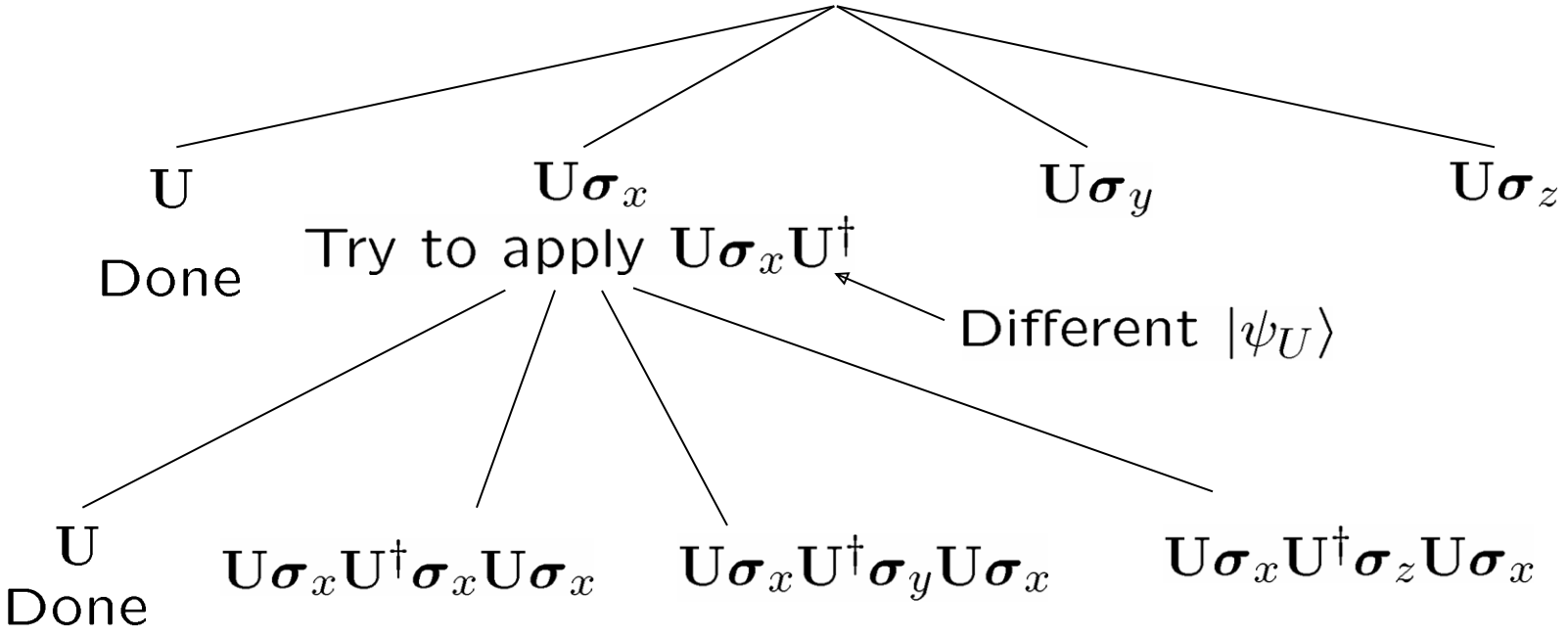
$$|\phi_U\rangle = \mathbf{I}\otimes\mathbf{U}\frac{1}{\sqrt{2}}(|01\rangle-|10\rangle)$$



$$|\phi_{U,i}\rangle = \mathbf{I}\otimes(\mathbf{U}\boldsymbol{\sigma}_i)\frac{1}{\sqrt{2}}(|01\rangle-|10\rangle)$$

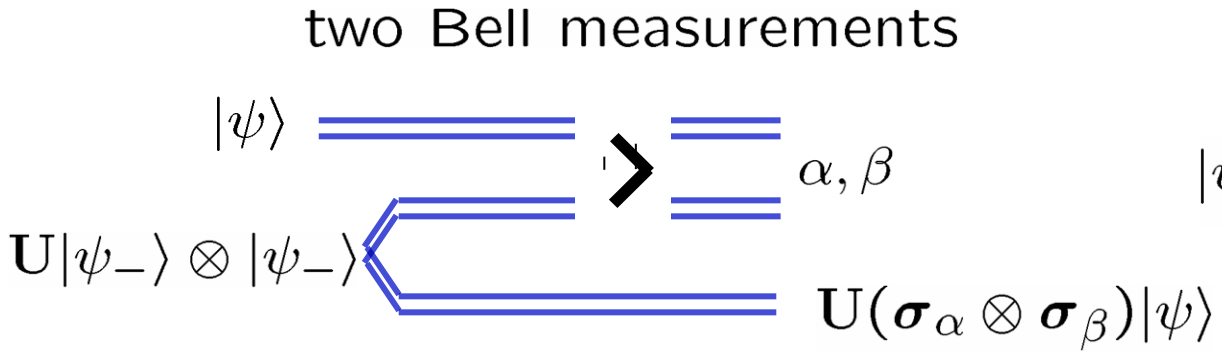
To prepare $|\phi_U\rangle$ repeatedly try $|\phi_{U,i}\rangle$ measurements.

If at First You Don't Succeed…



Exponentially small probability of failing after k levels.

Remote Controlling a NOT



$$|\psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Leung: $\mathbf{U}|\psi_-\rangle \otimes |\psi_-\rangle$ can be prepared using only 2 qubit projective measurements!

Single qubit rotations + CNOT = universal gate set

Food For Thought

Tonight's Menu

adiabatic quantum computer..... $\$\sqrt{\text{money in your wallet}}$ or $\$\infty$?

topological quantum computation..... ~$109

quantum computer with only measurements..... $0.37

Chez Quantum

For entry into literature see `http://arXiv.org/quant-ph`

Adiabatic: Farhi, Goldstone, Gutman, Childs, Roland, Cerf, Preskill, Ruskai, van Dam, Mosca, Vazirani, Hogg

Topological: Kitaev, Preskill, Freedman, Landahl, Wang

Measurement Based: Nielsen, Leung, Knill, Laflamme, Milburn, Raussendorf, Breigel